Recommendation Visualization Logic

AID-LOG-011 | Revision 1 | MVP Standard

Purpose. This standard defines how AIDOOi visually represents the real product compositions produced by AID-LOG-010. Visualization supports the recommendation; it does not replace product, fit, composition, budget or availability logic.

AIDOOi visualization is a visual representation of the recommendation - it must never become a substitute for the underlying product, fit, composition or availability logic.

1. Product Identity Preservation

AIDOOi must preserve the identity of the actual recommended products and must not replace them with AI-generated lookalikes.

Rule / Element	MVP Requirement
Preserve where reliably possible	Overall form, colour, visible material, legs/base, arms, major cushions/sections and key proportions/details.
MVP fidelity target	Recognizably product-faithful. The system does not claim 100% pixel-perfect reproduction.
Insufficient fidelity	If a product cannot be represented with sufficient fidelity, AIDOOi must not present a fabricated lookalike as an exact representation.

The customer should not open the underlying Product Card and discover a materially different real product.

2. Room Context

Visualization operates in two context modes depending on the customer input available.

Rule / Element	MVP Requirement
Customer-Space Visualization	When a room photo is available, preserve the recognizable room context as far as possible. Do not unnecessarily alter visible walls, windows, doors or fixed architectural elements. Insert the selected composition into that space.
Concept Visualization	When no customer room photo is available, AIDOOi may create a representative environment using known information such as room type, dimensions where available, Style and Colour Direction. It must not imply an exact reconstruction of the customer's home.
Floor plan	A floor plan may support spatial understanding, but by itself is not sufficient to claim an exact photorealistic reconstruction.

When customer-space imagery is available, AIDOOi should preserve the recognizable room context. When it is unavailable, AIDOOi may create a representative concept environment without implying that it reproduces the customer's actual space.

3. Product Scale & Proportion

Fit Logic and Visualization Scale are separate responsibilities. Fit determines suitability; visualization must avoid implausible relative sizing.

Rule / Element	MVP Requirement
Relative Product Scale - Required	Use structured Width, Depth and Height data to keep recommended products proportionally credible relative to each other.
Exact Room Placement - Conditional	Only imply measured placement when sufficient spatial data exists. A normal room photo without a reliable reference measurement is not enough for centimetre-accurate placement.
Concept boundary	Where spatial data is insufficient, the image remains conceptual even if the products themselves are product-faithful.

AIDOOi must preserve credible relative scale between recommended products using their structured dimensions, while exact spatial placement must not be implied where sufficient room measurements are unavailable.

4. Composition Integrity

The visualization must faithfully represent the composition produced by AID-LOG-010. AI may complete the environment visually, but must not silently redesign the recommendation.

Rule / Element	MVP Requirement
Recommended products	Anchor and Core recommended products must not be replaced, invented or omitted in a way that changes the recommendation.
Environmental context	AI may add minor non-commercial environmental details for realism, such as lighting or subtle styling context.
No misleading additions	Do not add furniture or prominent decor that could be mistaken for a purchasable part of the recommendation.
Lean compositions	Do not fill intentional empty space with additional furniture merely to make the image look fuller.
Quantity	Represented quantity should correspond to the composition, subject only to natural occlusion or camera framing.

Recommended products are locked. AI may add minor non-commercial environmental details for realism, but must not introduce furniture or prominent decor that could be mistaken for part of the recommendation.

5. Product Replacement & Visualization Update

A customer may replace an individual recommended product without restarting the journey. The replacement must be validated before the updated visualization becomes active.

Rule / Element	MVP Requirement
Replacement flow	Select Product -> Replace -> Show Suitable Alternatives -> Customer Selects -> Revalidate Composition -> Update Visualization.
Revalidation	Alternative must pass Hard Fit Requirements, AID-LOG-009 minimum Match Score, composition compatibility, budget impact and availability checks.
Price impact	Show the change in product price and updated room/project total immediately.
No silent redesign	Replacing one product must not automatically replace other products without the customer's knowledge. If another product becomes unsuitable, AIDOOi may recommend a further change.
Re-score	After replacement, recalculate the relevant AID-LOG-010 Composition Score.

A customer may replace an individual recommended product without restarting the journey, provided the replacement is revalidated against fit, match quality, composition compatibility, budget and availability.

6. Visualization Confidence & Disclosure

AIDOOi uses two simple visualization confidence levels for MVP.

Rule / Element	MVP Requirement
Product-Faithful Visualization	Used when product imagery and structured data support a recognizably faithful representation. It remains AI-generated and is not presented as pixel-perfect.
Concept Visualization	Used when information is insufficient for reliable product or space reproduction. The image communicates Design Direction and Composition rather than exact appearance.
Customer disclosure	Display a small, clear label indicating the applicable visualization level.

AIDOOi must never present a concept-level AI visualization as an exact representation of the customer's space or the recommended products.

7. Multi-Room Project Visualization

Full Apartment and Villa recommendations are project-level recommendations. Each AID-LOG-010 option may contain multiple coordinated room visualizations.

Rule / Element	MVP Requirement
Full Apartment	May contain coordinated visualizations for Bedroom module(s), Living Room, Dining Room and Home Office when selected.
Villa	May contain coordinated visualizations for multiple Bedrooms, Living Areas, Dining Areas and Home Office across relevant floors.
Project coherence	All room visualizations within one project option must respect Overall Style, Overall Colour Direction, Material Harmony and budget logic.
Repeated-space preference	Same Products / Different Products preferences from AID-SPC-007 and AID-SPC-008 must be respected wherever fit, function and availability allow.
Project meaning	The customer chooses between coordinated project recommendations, not unrelated room images.

For Full Apartment and Villa paths, one final recommendation is a coordinated project containing multiple room visualizations governed by the same project-level recommendation logic.

8. Visualization-to-Product Interaction

Every purchasable product presented as part of the recommendation must be traceable to the underlying real product and current seller offer.

Rule / Element	MVP Requirement
Interactive product access	A recommended product shown in the visualization should be selectable, for example through a tap/hotspot, to open its Product Card.
Product Card	At minimum show Product Name, Brand, Current Price, Availability, Viewing Status/Location and actions such as View Details and Replace.
Commercial traceability	Resolve from canonical AID Product ID to the applicable AIO Seller Offer so price, stock, viewing and delivery data come from the live offer rather than the image.
Hotspot boundary	Only recommended purchasable products receive Product Cards. Non-commercial environmental details do not.
Seller prominence	Seller information is available inside the Product Card but should not dominate the main visualization.

Rule / Element	MVP Requirement
Project summary	Multi-room recommendations should provide a project-level summary showing total current price and access to all included products.

Every purchasable product shown as part of an AIDOOi recommendation must be traceable from the visualization to its actual product record and current seller offer.

9. System Relationships

Standard	Role in Visualization
AID-LOG-009	Supplies eligible, ranked real products and minimum product Match Score.
AID-LOG-010	Supplies the final coordinated composition(s) that LOG-011 must represent faithfully.
AID-LOG-002 to AID-LOG-008	Supply room/project composition, fit, essential-function and budget context.
AID-DATA-005	Provides canonical AID Product ID and AIO Seller Offer traceability.
AID-DATA-006	Provides product dimensions used for credible relative scale.
AID-DATA-007 / 008 / 009	Provide Colour, Material and Style data supporting faithful and coherent representation.
AID-DATA-011	Provides availability, viewing and delivery status through the seller offer.
AID-DATA-012	Provides product imagery/media used to preserve product identity.
AID-SPC-007 / 008	Provide multi-room structure and Same Products / Different Products preferences.

10. MVP Boundary

Revision 1 deliberately avoids claims of unrestricted AI interior-design understanding, centimetre-accurate placement from ordinary photographs, or guaranteed pixel-perfect product reproduction. The MVP objective is a trustworthy, product-linked visual layer built on the structured recommendation system already defined by AIDOOi.

Final Principle: The image may improve understanding and confidence, but the recommendation remains defined by the real products, structured data, fit rules, composition logic, current seller offers and customer choices.